| Hands-on Activity 0.2 | |
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| **Hands-on Activity 0.2 - I/O read and write** | |
| **Course Code:** CPE 009B | **Program:** Computer Engineering |
| **Course Title:** Object Oriented Programming | **Date Performed: 07-18-26** |
| **Section: CPE21S2** | **-Date Submitted: 07-18-26** |
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**A. Instructions**

**Instructions:**

- Make a copy of the google docs for your documentation: Prelim_OOPB
- Download Prelim_OOPB
- and rename it as YourFullname_PrelimExamOOPB.doc or .pdf.
- Create a single Python application that acts as a simple Recipe Organizer. The application should allow a user to add new recipes, and view a collection of minimum of 10 recipes
- Submit your python file and name it **YourFullname_Prelim.py** your application should run
- Use comments as an explanation on the OOP design choices you made (Why you made a class, the function, etc, explain the OOP concept connected to the class created)
- At the very end of this document, you will post the code and the screenshot of the corresponding output to the code. For example, in step 1, you will create the code, then screenshot the corresponding output of the code.
- Do not forget to paste the Google colab link of your work below the output.
- NO AI ALLOWED, ANYONE FOUND CHEATING WILL FAIL AS THIS VIOLATES THE STUDENT HANDBOOK.

**B. Output(s) and Observation(s)**

```
class Recipe: # Creates the blueprint for making individual recipe objects.
  def __init__(self, repNam, ingredients, instructions): # Sets up the
initialization function to accept incoming recipe data.
    self.repNam = repNam # Saves the recipe name inside the object.
    self.ingredients = ingredients #Saves the ingredients list inside the object.
    self.instructions = instructions# Saves the instructions list inside the object.

  def display(self): #  Creates a function to print out the recipe details.
    print("\n----------Cook Book----------\n")#rints a clean, decorative top
border with spacing.
    print(f"{self.repNam.upper()}")# Prints the recipe name in all capital letters.
    print("Ingredients: ")#Prints the section title for ingredients.
    for idx, ingredient in enumerate(self.ingredients, 1):# Loops through the
ingredients list, starting the counter at 1.
      print(f"{idx}. {ingredient}") #Prints the current item number and the
ingredient name.
```

```
    print("\nIntructions:\n") # Prints the section title for instructions with
spacing.
    for idx, instruction in enumerate(self.instructions, 1):#Loops through the
instructions list, starting the counter at 1.
            print(f"Step {idx}: {instruction}") #Prints out the formatted step
number and the instruction.
```

```
class recipeOrganizer: #reates the blueprint for the organizer that will manage all
your recipes.
  def __init__(self): # Sets up the initialization function when a new organizer is
created.
    self.collection = [] # Creates an empty list inside the object to store your
recipes.

  def add_recipe(self, recipe): #Creates a function to add a new recipe object to
your organizer.
    self.collection.append(recipe) #Saves the new recipe into your collection list.
    print("\n-----------Recipe Organizer-----------") # Prints a decorative header
when a recipe is added.
    for idx, recipe in enumerate(self.collection, 1): #Loops through every recipe in
the collection, starting the count at 1.
      print(f"added '{recipe.repNam}' to the organizer.") # Prints a confirmation
message showing which recipe was added.

  def listAllRecipe(self): #Creates a function to display the names of all saved
recipes.
    if not self.collection:#Checks if your collection list is completely empty.
      print("The organizer is empty.") #ells the user there are no recipes to
display.
      return #Stops the function early so it does not try to print an empty list.

    print("\n-----------My Recipes-----------") # Prints a section title for your
saved recipe list.
    for idx, recipe in enumerate(self.collection, 1): #Loops through all your saved
recipes, starting the counter at 1.
      print(f"{idx}. {recipe.repNam}")# Prints a clean, numbered list of your recipe
names.
```

```
# main.py - this is where the code starts and goes through the classes.
#There are the names, the recipes, the organizer, the instructions and ingredients

if __name__ == "__main__":
    my_organizer = recipeOrganizer()
```

```
recipe1 = Recipe(
    "Pancakes",
    ["Flour", "Butter", "Egg", "Milk", "Salt"],

     ("Get the ingredients in a bowl.",
     "Mix.",
     "Fry on a pan.")
)

recipe2 = Recipe(
    "Bacon and Rice",
    ["Bacon", "Oil", "Rice"],

     ("Defreeze the get-ready-to-eat bacon",
      "Cook the rice.",
      "Fry the bacon on the pan.")
)

recipe3 = Recipe(
    "Adobo",
    ["Meat", "Soy Sauce", "Dried bay leaves",
     "Vinegar", "Garlic", "Whole peppercorns",
     "Salt", "Water"],

     ("The pork marinates first",
     "The pork sears in its own marinade.",
     "Simmer with Aromatics.",
     "Finish with Vinegar")
)

recipe4 = Recipe(
    "Air Fryer Buffalo Wings",
    ["Chicken wings",
     "Unsalted butter ",
     "Frank's RedHot"],

     ("Prepare the Wings",
     "Air Fry the Wings.",
     "Make the Sauce and Coat")
)

recipe5 = Recipe(
    "Prosciutto Wrapped Cantaloupe",
```

```
        ["1 small cantaloupe, peeled, deseeded & sliced into 1/2 inch slices",
         "12 ounces thinly sliced prosciutto",
         "1/4 cup balsamic glaze or reduced balsamic "],

         ("Gently wrap one slice of prosciutto around the middle of each slice of
cantaloupe.",
         "I suggest tearing the prosciutto in half lengthwise",
         "wrapping each half around the cantaloupe spear to cover more of the
cantaloupe's surface.",
         "Drizzle the balsamic glaze over the prosciutto wrapped cantaloupe.",
         "Transfer to a serving dish & enjoy immediately.")
    )

    recipe6 = Recipe(
        "Chocolate and Strawberry Energy Bites",
        ["1 cup Simple Truth Organic 100% Whole Grain Rolled Oats",
         "½ cup Simple Truth Organic Crunchy Peanut Butter",
         "⅓ cup Simple Truth Organic Whole Ground Flaxseed Meal",
         "¼ cup Simple Truth Organic Honey",
         "1 tablespoon water",
         "⅓ cup Simple Truth Gluten-Free Freeze-Dried Strawberries, crumbled",
         "⅓ cup Simple Truth Organic Semi-Sweet Chocolate Chips"],

         ("In a large mixing bowl",
         "Combine the oats, Simple Truth Organic Crunchy Peanut Butter, flaxseed,
honey and water.",
         "Fold in the Simple Truth Freeze-Dried Strawberries and chocolate chips.",
         "Cover with plastic wrap and refrigerate for 30 minutes.",
         "Using clean hands, roll the mixture into approximately 1-inch balls before
serving. ",
         "The energy bites can be stored in an airtight container in the
refrigerator for up to a week.")
    )

    recipe7 = Recipe(
        "Apple Pie Pizza",
        ["1 (14 ounce) package refrigerated pie crust",
         "6 tablespoons softened butter, divided",
         "4 medium Fuji or Granny Smith apples, peeled, cored, and cut into 1-inch
pieces",
         "1/4 teaspoon kosher salt, divided",
         "1 1/2 teaspoons ground cinnamon, divided",
         "1/2 cup white sugar",
         "1/3 cup cold water, divided",
```

```
        "1 tablespoon cornstarch",
        "1/2 cup brown sugar",
        "1/2 cup all-purpose flour",
        "2/3 cup shredded white Cheddar cheese",
        "1 cup confectioner’s sugar",
        "2 tablespoons milk"],

       ("Remove pie crusts from the refrigerator and let stand at room temperature
while filling is prepared.",
        "Heat 2 tablespoons butter in a large skillet until melted.",
        "Add apples",
        "1/8 teaspoon salt",
        "1 1/4 teaspoon cinnamon",
        "white sugar",
        "3 tablespoons of water",
        "Stir until well combined and bring mixture to a simmer.",
        "Cook, stirring occasionally, until apples are tender, about 8 to 10
minutes.",
        "Meanwhile, whisk together remaining water and cornstarch until it is
dissolved.",
        "Add cornstarch mixture to apple mixture and stir until well
incorporated.",
        "Bring mixture to a simmer and cook one minute until thickened. ",
        "Remove from heat and let cool.",
        "Meanwhile add flour and brown sugar to a bowl.",
        "Add in remaining butter, salt, and cinnamon and mix with hands or a fork
until well combined and crumbly. Set aside.",
        "Preheat the oven to 400 degrees F (200 degrees C).",
        "Stack 2 pie crusts on top of each other and press them together.",
        "Use a rolling pin to roll the crust out to 12 to 14 inches and to form one
crust.",
        "Transfer dough to a pizza pan and, If desired, crimp the edges like a
pie.",
        "Sprinkle crust with cheese and top with apple mixture.",
        "Add brown sugar crumble evenly to the top.",
        "Bake in the preheated oven until browned and bubbly, 22 to 25 minutes.",
        "Let stand at least 10 minutes before serving.",
        "Whisk confectioner’s sugar and milk together and drizzle mixture over the
“pizza.”  ",
        "Cut into wedges and serve.")
    )

    recipe8 = Recipe(
        "Oatmeal Creme Pie Ice Cream Sandwiches",
```

```
        ["1/4 cup unsalted butter, softened",
         "1/2 cup packed light brown sugar",
         "1 large egg yolk",
         "1 teaspoon vanilla extract",
         "1 teaspoon unsulphured molasses (optional)",
         "1/2 cup old fashioned oats",
         "1/2 cup oat flour",
         "1/2 cup oat flour",
         "1/2 teaspoon baking soda",
         "1/4 teaspoon baking powder",
         "1/4 teaspoon salt",
         "1/2 teaspoon ground cinnamon",
         "1/8 teaspoon ground nutmeg",
         "2 pints vanilla ice cream"],

         ("Preheat the oven to 350 degrees F (180 degrees C). ",
         "Line a baking sheet with parchment paper.",
         "Beat butter and brown sugar together with an electric mixer in a large
bowl until light and fluffy, 2 to 3 minutes.",
         "Beat in egg yolk, vanilla, and molasses until smooth.",
         "In a separate bowl, whisk together oats, oat flour, cornstarch, baking
soda, baking powder, salt, cinnamon, and nutmeg.",
         "Fold oat mixture into butter mixture until just combined. Do not
overmix.",
         "Let dough rest for 15 to 20 minutes to allow the oat flour and oats to
fully hydrate.",
         "Scoop about 1 1/2 tablespoons dough for each cookie and roll into balls.",
         "Place cookies about 2 inches apart on the prepared baking sheet and gently
flatten them to about 2 1/2 inches in diameter.",
         "Bake in the preheated oven just until the edges are set and the centers
still look slightly underbaked, 9 to 11 minutes.",
         "Let cookies cool on the baking sheet for 10 minutes, then transfer  to a
wire rack to cool completely.",
         "For ice cream sandwiches, freeze cooled cookies until firm, 20 to 30
minutes, before assembling. ",
         "Add 1 scoop, or about 1/4 cup, slightly softened ice cream between 2
cookies.",
         "Press gently, wrap each sandwich tightly, and freeze for at least 4
hours.")
    )

    recipe9 = Recipe(
        "Affogato",
        ["2 tsp NESCAFÉ® Ice Roast",
```

```
        "2 scoops vanilla ice cream ",
        "chocolate shavings (optional)",
        "pistachios (optional)"],

        ("Mixing jug",
        "Put it in a 250 ml serving bowl",
        "Then put cream on top")
    )

    recipe10 = Recipe(
        "Café Frappé with Honey",
        ["2 tsp NESCAFÉ® Ice Roast",
        "60 ml cold water ",
        "15 ml honey",
        "200 ml milk",
        "ice cubes"],

        ("300 ml glass",
        "Shaker or mason jar")
    )




    my_organizer.add_recipe(recipe1)
    my_organizer.add_recipe(recipe2)
    my_organizer.add_recipe(recipe3)
    my_organizer.add_recipe(recipe4)
    my_organizer.add_recipe(recipe5)
    my_organizer.add_recipe(recipe6)
    my_organizer.add_recipe(recipe7)
    my_organizer.add_recipe(recipe8)
    my_organizer.add_recipe(recipe9)
    my_organizer.add_recipe(recipe10)

    my_organizer.listAllRecipe()

    recipe1.display()
    recipe2.display()
    recipe3.display()
    recipe4.display()
    recipe5.display()
    recipe6.display()
```

```
    recipe7.display()
    recipe8.display()
    recipe9.display()
    recipe10.display()
```

```
    ----------Recipe Organizer----------
    added 'Pancakes' to the organizer.

    ----------Recipe Organizer----------
    added 'Pancakes' to the organizer.
    added 'Bacon and Rice' to the organizer.

    ----------Recipe Organizer----------
    added 'Pancakes' to the organizer.
    added 'Bacon and Rice' to the organizer.
    added 'Adobo' to the organizer.

    ----------Recipe Organizer----------
    added 'Pancakes' to the organizer.
    added 'Bacon and Rice' to the organizer.
    added 'Adobo' to the organizer.
    added 'Air Fryer Buffalo Wings' to the organizer.

    ----------Recipe Organizer----------
    added 'Pancakes' to the organizer.
    added 'Bacon and Rice' to the organizer.
    added 'Adobo' to the organizer.
    added 'Air Fryer Buffalo Wings' to the organizer.
    added 'Prosciutto Wrapped Cantaloupe' to the organizer.

    ----------Recipe Organizer----------
    added 'Pancakes' to the organizer.
    added 'Bacon and Rice' to the organizer.
    added 'Adobo' to the organizer.
    added 'Air Fryer Buffalo Wings' to the organizer.
    added 'Prosciutto Wrapped Cantaloupe' to the organizer.
    added 'Chocolate and Strawberry Energy Bites' to the organizer.
```

```
----------Recipe Organizer----------
added 'Pancakes' to the organizer.
added 'Bacon and Rice' to the organizer.
added 'Adobo' to the organizer.
added 'Air Fryer Buffalo Wings' to the organizer.
added 'Prosciutto Wrapped Cantaloupe' to the organizer.
added 'Chocolate and Strawberry Energy Bites' to the organizer.
added 'Apple Pie Pizza' to the organizer.

----------Recipe Organizer----------
added 'Pancakes' to the organizer.
added 'Bacon and Rice' to the organizer.
added 'Adobo' to the organizer.
added 'Air Fryer Buffalo Wings' to the organizer.
added 'Prosciutto Wrapped Cantaloupe' to the organizer.
added 'Chocolate and Strawberry Energy Bites' to the organizer.
added 'Apple Pie Pizza' to the organizer.
added 'Oatmeal Creme Pie Ice Cream Sandwiches' to the organizer.

----------Recipe Organizer----------
added 'Pancakes' to the organizer.
added 'Bacon and Rice' to the organizer.
added 'Adobo' to the organizer.
added 'Air Fryer Buffalo Wings' to the organizer.
added 'Prosciutto Wrapped Cantaloupe' to the organizer.
added 'Chocolate and Strawberry Energy Bites' to the organizer.
added 'Apple Pie Pizza' to the organizer.
added 'Oatmeal Creme Pie Ice Cream Sandwiches' to the organizer.
added 'Affogato' to the organizer.
```

```
----------Recipe Organizer----------
added 'Pancakes' to the organizer.
added 'Bacon and Rice' to the organizer.
added 'Adobo' to the organizer.
added 'Air Fryer Buffalo Wings' to the organizer.
added 'Prosciutto Wrapped Cantaloupe' to the organizer.
added 'Chocolate and Strawberry Energy Bites' to the organizer.
added 'Apple Pie Pizza' to the organizer.
added 'Oatmeal Creme Pie Ice Cream Sandwiches' to the organizer.
added 'Affogato' to the organizer.
added 'Café Frappé with Honey' to the organizer.

----------My Recipes----------
1. Pancakes
2. Bacon and Rice
3. Adobo
4. Air Fryer Buffalo Wings
5. Prosciutto Wrapped Cantaloupe
6. Chocolate and Strawberry Energy Bites
7. Apple Pie Pizza
8. Oatmeal Creme Pie Ice Cream Sandwiches
9. Affogato
10. Café Frappé with Honey

----------Cook Book----------

PANCAKES
Ingredients:
1. Flour
2. Butter
3. Egg
4. Milk
5. Salt
```

```
    Intructions:

    Step 1: Get the ingredients in a bowl.
    Step 2: Mix.
    Step 3: Fry on a pan.

    -----------Cook Book-----------

    BACON AND RICE
    Ingredients:
    1. Bacon
    2. Oil
    3. Rice

    Intructions:

    Step 1: Defreeze the get-ready-to-eat bacon
    Step 2: Cook the rice.
    Step 3: Fry the bacon on the pan.

    -----------Cook Book-----------

    ADOBO
    Ingredients:
    1. Meat
    2. Soy Sauce
    3. Dried bay leaves
    4. Vinegar
    5. Garlic
    6. Whole peppercorns
    7. Salt
    8. Water
```

```
    Intructions:

    Step 1: The pork marinates first
    Step 2: The pork sears in its own marinade.
    Step 3: Simmer with Aromatics.
    Step 4: Finish with Vinegar

    -----------Cook Book-----------

    AIR FRYER BUFFALO WINGS
    Ingredients:
    1. Chicken wings
    2. Unsalted butter
    3. Frank's RedHot

    Intructions:

    Step 1: Prepare the Wings
    Step 2: Air Fry the Wings.
    Step 3: Make the Sauce and Coat

    -----------Cook Book-----------

    PROSCIUTTO WRAPPED CANTALOUPE
    Ingredients:
    1. 1 small cantaloupe, peeled, deseeded & sliced into 1/2 inch slices
    2. 12 ounces thinly sliced prosciutto
    3. 1/4 cup balsamic glaze or reduced balsamic
```

```
Intructions:

Step 1: Gently wrap one slice of prosciutto around the middle of each slice of cantaloupe.
Step 2: I suggest tearing the prosciutto in half lengthwise
Step 3: wrapping each half around the cantaloupe spear to cover more of the cantaloupe's surface.
Step 4: Drizzle the balsamic glaze over the prosciutto wrapped cantaloupe.
Step 5: Transfer to a serving dish & enjoy immediately.

-----------Cook Book-----------

CHOCOLATE AND STRAWBERRY ENERGY BITES
Ingredients:
1. 1 cup Simple Truth Organic 100% Whole Grain Rolled Oats
2. ½ cup Simple Truth Organic Crunchy Peanut Butter
3. ⅓ cup Simple Truth Organic Whole Ground Flaxseed Meal
4. ¼ cup Simple Truth Organic Honey
5. 1 tablespoon water
6. ⅓ cup Simple Truth Gluten-Free Freeze-Dried Strawberries, crumbled
7. ⅓ cup Simple Truth Organic Semi-Sweet Chocolate Chips
```

```
Intructions:

Step 1: In a large mixing bowl
Step 2: Combine the oats, Simple Truth Organic Crunchy Peanut Butter, flaxseed, honey and water.
Step 3: Fold in the Simple Truth Freeze-Dried Strawberries and chocolate chips.
Step 4: Cover with plastic wrap and refrigerate for 30 minutes.
Step 5: Using clean hands, roll the mixture into approximately 1-inch balls before serving.
Step 6: The energy bites can be stored in an airtight container in the refrigerator for up to a week.

-----------Cook Book-----------

APPLE PIE PIZZA
Ingredients:
1. 1 (14 ounce) package refrigerated pie crust
2. 6 tablespoons softened butter, divided
3. 4 medium Fuji or Granny Smith apples, peeled, cored, and cut into 1-inch pieces
4. 1/4 teaspoon kosher salt, divided
5. 1 1/2 teaspoons ground cinnamon, divided
6. 1/2 cup white sugar
7. 1/3 cup cold water, divided
8. 1 tablespoon cornstarch
9. 1/2 cup brown sugar
10. 1/2 cup all-purpose flour
11. 2/3 cup shredded white Cheddar cheese
12. 1 cup confectioner's sugar
13. 2 tablespoons milk
```

```
Intructions:

Step 1: Remove pie crusts from the refrigerator and let stand at room temperature while filling is prepared.
Step 2: Heat 2 tablespoons butter in a large skillet until melted.
Step 3: Add apples
Step 4: 1/8 teaspoon salt
Step 5: 1 1/4 teaspoon cinnamon
Step 6: white sugar
Step 7: 3 tablespoons of water
Step 8: Stir until well combined and bring mixture to a simmer.
Step 9: Cook, stirring occasionally, until apples are tender, about 8 to 10 minutes.
Step 10: Meanwhile, whisk together remaining water and cornstarch until it is dissolved.
Step 11: Add cornstarch mixture to apple mixture and stir until well incorporated.
Step 12: Bring mixture to a simmer and cook one minute until thickened.
Step 13: Remove from heat and let cool.
Step 14: Meanwhile add flour and brown sugar to a bowl.
Step 15: Add in remaining butter, salt, and cinnamon and mix with hands or a fork until well combined and crumbly.
Step 16: Preheat the oven to 400 degrees F (200 degrees C).
Step 17: Stack 2 pie crusts on top of each other and press them together.
Step 18: Use a rolling pin to roll the crust out to 12 to 14 inches and to form one crust.
Step 19: Transfer dough to a pizza pan and, If desired, crimp the edges like a pie.
Step 20: Sprinkle crust with cheese and top with apple mixture.
Step 21: Add brown sugar crumble evenly to the top.
Step 22: Bake in the preheated oven until browned and bubbly, 22 to 25 minutes.
Step 23: Let stand at least 10 minutes before serving.
Step 24: Whisk confectioner’s sugar and milk together and drizzle mixture over the “pizza.”
Step 25: Cut into wedges and serve.
```

```
----------Cook Book----------

OATMEAL CREME PIE ICE CREAM SANDWICHES
Ingredients:
1. 1/4 cup unsalted butter, softened
2. 1/2 cup packed light brown sugar
3. 1 large egg yolk
4. 1 teaspoon vanilla extract
5. 1 teaspoon unsulphured molasses (optional)
6. 1/2 cup old fashioned oats
7. 1/2 cup oat flour
8. 1/2 cup oat flour
9. 1/2 teaspoon baking soda
10. 1/4 teaspoon baking powder
11. 1/4 teaspoon salt
12. 1/2 teaspoon ground cinnamon
13. 1/8 teaspoon ground nutmeg
14. 2 pints vanilla ice cream
```

```
Intructions:

Step 1: Preheat the oven to 350 degrees F (180 degrees C).
Step 2: Line a baking sheet with parchment paper.
Step 3: Beat butter and brown sugar together with an electric mixer in a large bowl until light and fluffy, 2 to 3 minutes.
Step 4: Beat in egg yolk, vanilla, and molasses until smooth.
Step 5: In a separate bowl, whisk together oats, oat flour, cornstarch, baking soda, baking powder, salt, cinnamon, and nutmeg.
Step 6: Fold oat mixture into butter mixture until just combined. Do not overmix.
Step 7: Let dough rest for 15 to 20 minutes to allow the oat flour and oats to fully hydrate.
Step 8: Scoop about 1 1/2 tablespoons dough for each cookie and roll into balls.
Step 9: Place cookies about 2 inches apart on the prepared baking sheet and gently flatten them to about 2 1/2 inches in diameter.
Step 10: Bake in the preheated oven just until the edges are set and the centers still look slightly underbaked, 9 to 11 minutes.
Step 11: Let cookies cool on the baking sheet for 10 minutes, then transfer  to a wire rack to cool completely.
Step 12: For ice cream sandwiches, freeze cooled cookies until firm, 20 to 30 minutes, before assembling.
Step 13: Add 1 scoop, or about 1/4 cup, slightly softened ice cream between 2 cookies.
Step 14: Press gently, wrap each sandwich tightly, and freeze for at least 4 hours.

----------Cook Book----------

AFFOGATO
Ingredients:
1. 2 tsp NESCAFÉ® Ice Roast
2. 2 scoops vanilla ice cream
3. chocolate shavings (optional)
4. pistachios (optional)

Intructions:

Step 1: Mixing jug
Step 2: Put it in a 250 ml serving bowl
Step 3: Then put cream on top
```

```
----------Cook Book----------

CAFÉ FRAPPÉ WITH HONEY
Ingredients:
1. 2 tsp NESCAFÉ® Ice Roast
2. 60 ml cold water
3. 15 ml honey
4. 200 ml milk
5. ice cubes

Intructions:

Step 1: 300 ml glass
Step 2: Shaker or mason jar
```

**Google Colab Link:**
https://colab.research.google.com/drive/1nmuaUiOg43xCUx7YCnFBylRvnnTTkwj-?usp=sharing

**C. Conclusion & Lessons Learned**

In my conclusion in doing my prelims exam is easy but also hard, it is hard for me to do some of the code, but I do understand it, I just searched and searched how do a list, then I tried to recall some from the activities, and there I was able to apply what I have learned, searched and my knowledge. I used two classes, one is for the recipe, and one for the organizer, I wasn't so sure if I needed to list the recipes of one dish, but I did. So I was able to do 2 dishes for breakfast, lunch, the snacks and the beverages. Then I added comments per line to make myself understand what the code do, it's a bit rewritten by gemini, because the one I did wasn't properly done, it was only to make me understand, so I sent it to grammarly to fix it, so it's more appropriate if I passes it. After I did the classes, I had some trouble in printing it, it was so messy, so I had to set the classes again to make it more understandable, but due to my lack of practice, I wasn't able to

bring it to more a appropriately visually good, because the recipes are really long and many, it is very long. Nevertheless, I did try hard to make it good, and to follow the instructions properly.